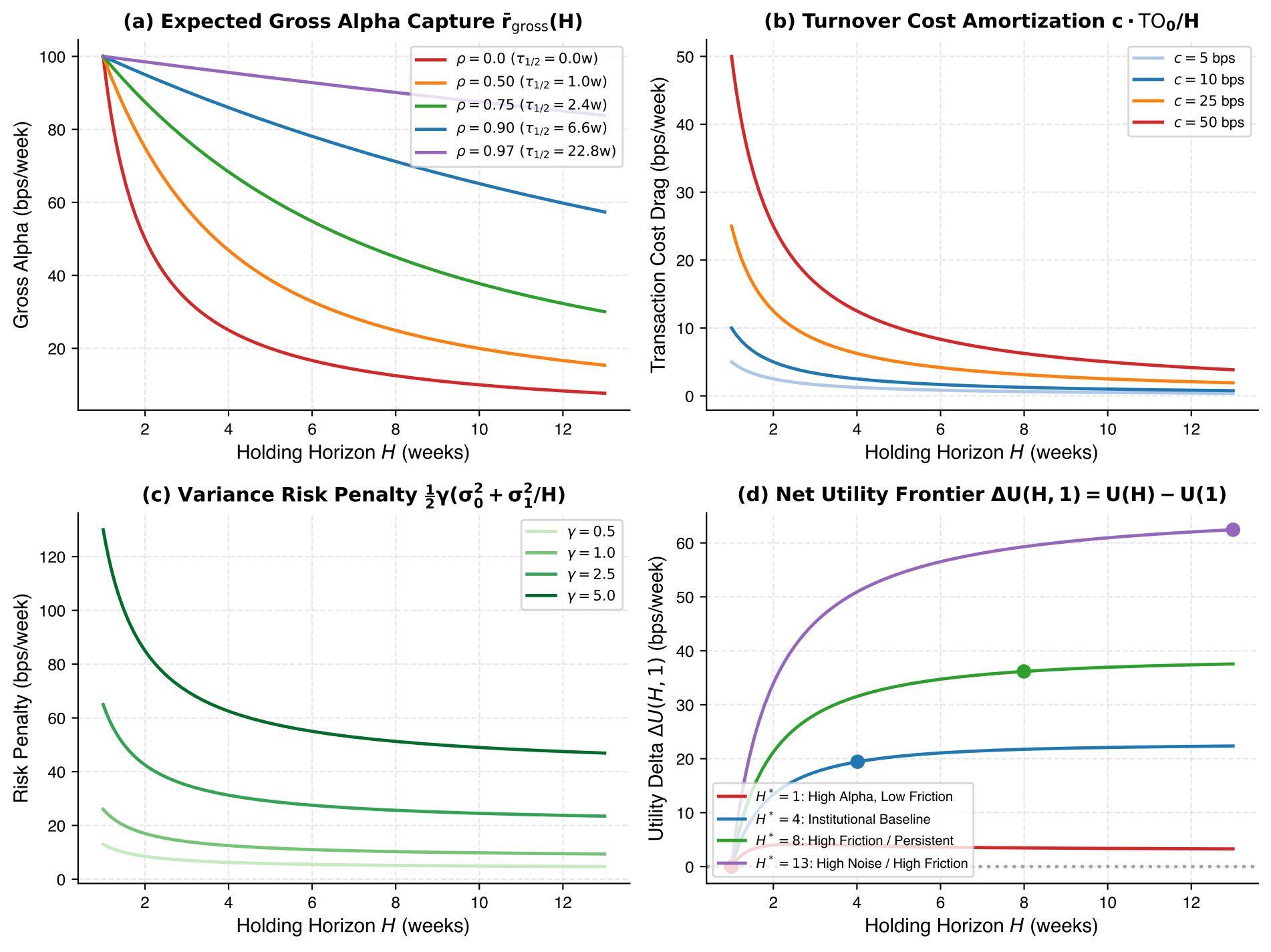

(a) Expected Gross Alpha Capture $\bar{r}_{gross}(H)$
Gross Alpha (bps/week)
Holding Horizon H (weeks)
$\rho = 0.0$ ($\tau_{1/2} = 0.0$w)
$\rho = 0.50$ ($\tau_{1/2} = 1.0$w)
$\rho = 0.75$ ($\tau_{1/2} = 2.4$w)
$\rho = 0.90$ ($\tau_{1/2} = 6.6$w)
$\rho = 0.97$ ($\tau_{1/2} = 22.8$w)
(b) Turnover Cost Amortization $c \cdot TO_0/H$
Transaction Cost Drag (bps/week)
Holding Horizon H (weeks)
$c = 5$ bps
$c = 10$ bps
$c = 25$ bps
$c = 50$ bps
(c) Variance Risk Penalty $\frac{1}{2}\gamma(\sigma_0^2 + \sigma_1^2/H)$
Risk Penalty (bps/week)
Holding Horizon H (weeks)
$\gamma = 0.5$
$\gamma = 1.0$
$\gamma = 2.5$
$\gamma = 5.0$
(d) Net Utility Frontier $\Delta U(H, 1) = U(H) - U(1)$
Utility Delta $\Delta U(H, 1)$ (bps/week)
Holding Horizon H (weeks)
$H^* = 1$: High Alpha, Low Friction
$H^* = 4$: Institutional Baseline
$H^* = 8$: High Friction / Persistent
$H^* = 13$: High Noise / High Friction